Example 10.20

Solve $y^2 + x^2 \frac{dy}{dx} = xy \frac{dy}{dx}$.

Solution

The given equation is rewritten as $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$.

This is a homogeneous differential equation.

Put $y = vx$. Then, we have $x\frac{dv}{dx} = \frac{v}{v-1}$.

By separating the variables, $\frac{v-1}{v} dv = \frac{dx}{x}$.

Integrating, we obtain $v - \log|v| = \log|x| + \log|C|$ or $v = \log|vxC|$.

Replacing v by $\frac{y}{x}$, we get, $\frac{y}{x} = \log|Cy|$ or $|Cy| = e^{y/x}$ or $y = ke^{y/x}$ (how!) which is the required solution.

Example 10.21

Solve $\left(1 + 2e^{x/y}\right) dx + 2e^{x/y}\left(1 - \frac{x}{y}\right) dy = 0$.

Solution

The given equation can be written as $\frac{dx}{dy} = \frac{\left(\frac{x}{y} - 1\right) 2e^{x/y}}{1 + 2e^{x/y}} = g\left(\frac{x}{y}\right)$. …(1)

The appearance of $\frac{x}{y}$ in equation (1), suggests that the appropriate substitution is $x = vy$.

Put $x = vy$. Then, we have $y\frac{dv}{dy} = -\frac{2e^v + v}{1 + 2e^v}$.

By separating the variables, we have $\frac{1 + 2e^v}{v + 2e^v} dv = -\frac{dy}{y}$.

On integration, we obtain

$\log|2e^v + v| = -\log|y| + \log|C|$ or $\log|2ye^v + vy| = \log|C|$ or $2ye^v + vy = \pm C$.

Replace v by $\frac{x}{y}$ to get, $2ye^{x/y} + x = k$, where $k = \pm C$, which gives the required solution.

EXERCISE 10.6

Solve the following differential equations:

1. $\left[x + y\cos\left(\frac{y}{x}\right)\right] dx = x\cos\left(\frac{y}{x}\right) dy$

2. $\left(x^3 + y^3\right) dy - x^2 y dx = 0$

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